MSE 360: Kinetic Processes in Materials
S2023
Homework 3
Due: January 28, 2023

Instructions: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

1. Answer true or false (and why):

   i. For the reaction A + 2B → C, the rate equation must be: $\frac{dC_A}{dt} = -kC_AC_B^2$.

   ii. Decreasing the activation energy of a reaction by half will increase the reaction rate by a factor of 2.

i) F rate eqn. not determined from strichiometry. Need experiment.

ii) F $k = k_0 \exp(-\Delta G_{act}/RT)$
Rxn rate (dC/dt) is exponentially dependent on ΔG_{act}.

2. Consider the integrated second-order rate law for a reaction which is first order with respect to two distinct reactants (A and B). Prove that the following expressions (Eqns. 3.52 and 3.53 in the textbook) are equivalent:

$$1) \; C_A = \frac{C_{B_0} - \frac{b}{a} C_{A_0}}{\frac{C_{B_0}}{C_{A_0}} \exp\left[-kt\left(\frac{b}{a} C_{A_0} - C_{B_0}\right)\right] - \frac{b}{a}} \; ; \quad 2) \; \frac{C_A}{C_B} = \frac{C_{A_0}}{C_{B_0}} exp\left[kt\left(\frac{b}{a} C_{A_0} - C_{B_0}\right)\right]$$

not dependent on C_B

Solve by substituting in Eqn. 1 for C_{B0} to get $\frac{C_A}{C_B}$ (1 substitution)

$$C_B = C_{B0} + \frac{b}{a}(C_A - C_{A0}) \qquad \text{Eqn. 3.46}$$

$$C_{B0} = C_B - \frac{b}{a}(C_A - C_{A0})$$

$$1) \; C_A = \frac{C_{B0} - \frac{b}{a} C_{A0}}{\frac{C_{B0}}{C_{A0}} \exp\left[-kt\left(\frac{b}{a} C_{A0} - C_{B0}\right)\right] - \frac{b}{a}}$$

$$C_A = \frac{\left[C_B - \frac{b}{a}(C_A - C_{A0})\right] - \frac{b}{a} C_{A0}}{\frac{C_{B0}}{C_{A0}} \exp\left[-kt\left(\frac{b}{a} C_{A0} - C_{B0}\right)\right] - \frac{b}{a}}$$

$$C_A = \frac{C_B - \frac{b}{a} C_A + \cancel{\frac{b}{a} C_{A0}} - \cancel{\frac{b}{a} C_{A0}}}{\frac{C_{B0}}{C_{A0}} \exp\left[-kt\left(\frac{b}{a} C_{A0} - C_{B0}\right)\right] - \frac{b}{a}}$$

$$C_A\left[\frac{C_{B0}}{C_{A0}}\exp\left(-kt\left(\frac{b}{a}C_{A0}-C_{B0}\right)\right)-\frac{b}{a}\right] = C_B - \frac{b}{a}C_A$$

$$\frac{C_A C_{B0}}{C_{A0}}\exp\left[-kt\left(\frac{b}{a}C_{A0}-C_{B0}\right)\right] - \cancel{\frac{b}{a}C_A} = C_B - \cancel{\frac{b}{a}C_A}$$

2 $$\boxed{\frac{C_A}{C_B} = \frac{C_{A0}}{C_{B0}}\exp\left[kt\left(\frac{b}{a}C_{A0}-C_{B0}\right)\right]}$$

3. If the rate constant k for a first-order reaction doubles with a change in temperature from T = 300 K to T = 320 K, what is ΔG_{act}?

Set up ratio of k_1/k_2 & solve for ΔG_{act}

$k_2 = 2k_1$

$$\frac{k_1}{k_2} = \frac{\cancel{k_0}\exp(-\Delta G_{act}/RT_1)}{\cancel{k_0}\exp(-\Delta G_{act}/RT_2)}$$

$$\frac{k_1}{k_2} = \exp\left[-\frac{\Delta G_{act}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right]$$

$$\Delta G_{act} = \frac{-R\ln\frac{k_1}{k_2}}{\frac{1}{T_1} - \frac{1}{T_2}} \quad \rightarrow k_2 = 2k_1$$

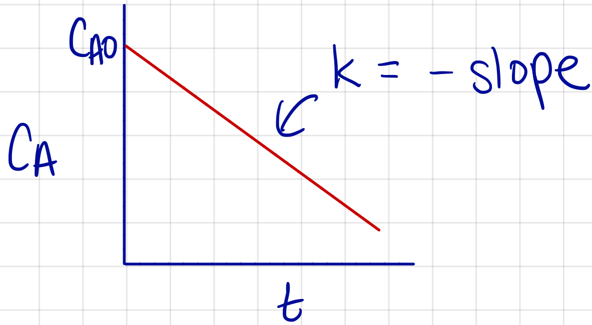
$$= \frac{-8.314\ J/mol \cdot K \ \ln 0.5}{\frac{1}{300K} - \frac{1}{320K}}$$

$$\boxed{\Delta G_{act} = 27.7\ kJ/mol}$$

4. For a 0, 1st, and 2nd-order rate reaction with respect to the concentration of reactant A, explain how to determine k, the rate constant, from data of the concentration of A vs. time. *Use graphs to support your explanation.*

0 order

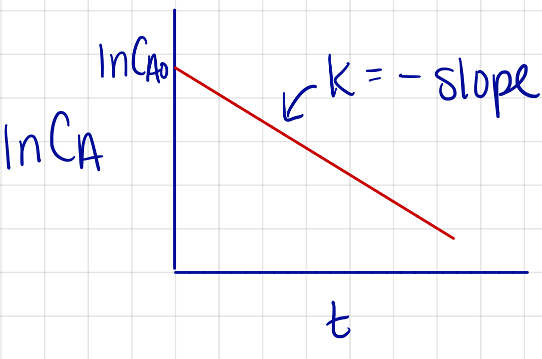
$C_A = C_{A0} - kt \Rightarrow$ plot of C_A vs. t is linear



1st order

$C_A = C_{A0} \exp(-kt)$

$\ln C_A = \ln C_{A0} - kt \Rightarrow$ plot of $\ln C_A$ vs. t is linear



2nd - order

$$\frac{1}{C_A} = \frac{1}{C_{A0}} + kt \Rightarrow$$ plot of $\frac{1}{C_A}$ vs. t is linear